Monocyte-endothelial cell interactions heightened in clonal hematopoiesis of indeterminate potential

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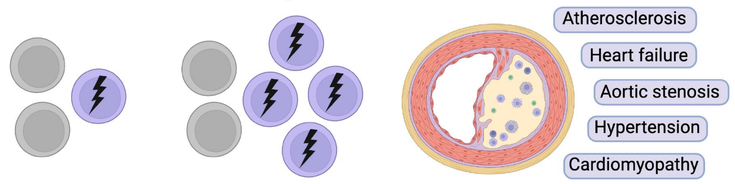
INTRODUCTION

Hematopoietic stem cells can incur mutations...

...that lead to proliferative advantages...

...associated with vascular diseases[1].



Driver genes are lowly expressed in blood...

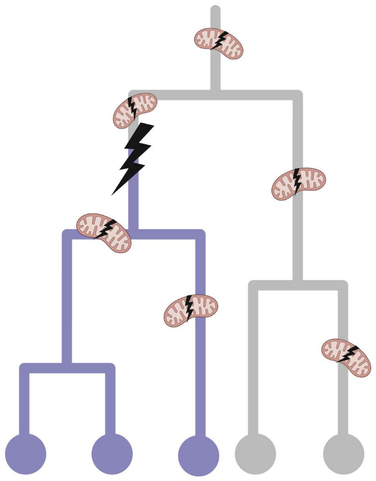
...making **mutant** and **wildtype** cells difficult to distinguish.

STUDY DESIGN

Goal: Characterize the effects of clonal hematopoiesis of indeterminate potential (CHIP) on monocyte-endothelial signaling patterns.

Mitochondrial genes are...

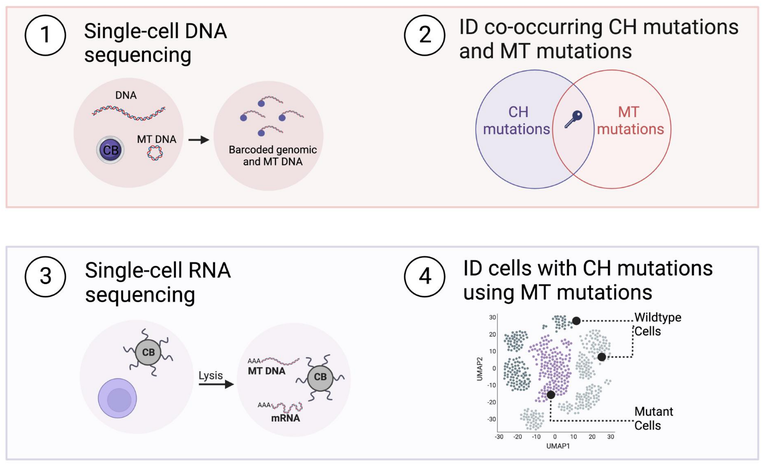
1. Highly expressed
2. Ubiquitously expressed
3. Frequently mutated



We performed single-cell RNA sequencing on peripheral blood mononuclear cells and subcutaneous adipose tissue from 6 CHIP patients and 6 matched controls. We used mitochondrial mutations to identify cells carrying CHIP mutations.

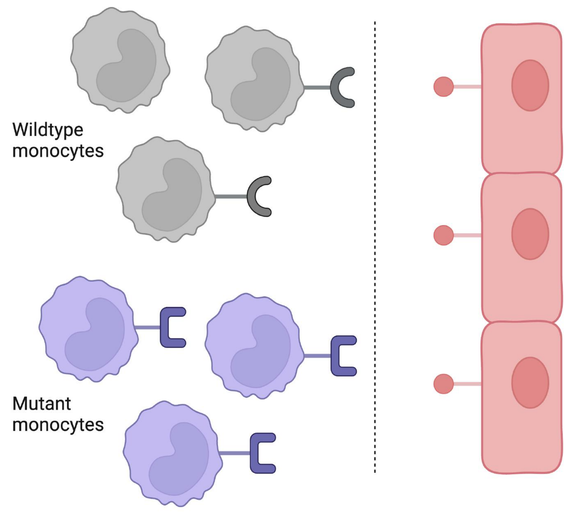
We validated our findings using publicly available data from Rauch *et al.*'s *TET2* knockout mouse study[2] and from the Chan Zuckerberg Initiative's CELLxGENE database[3].

Mitochondrial lineage tracing methodology[4]

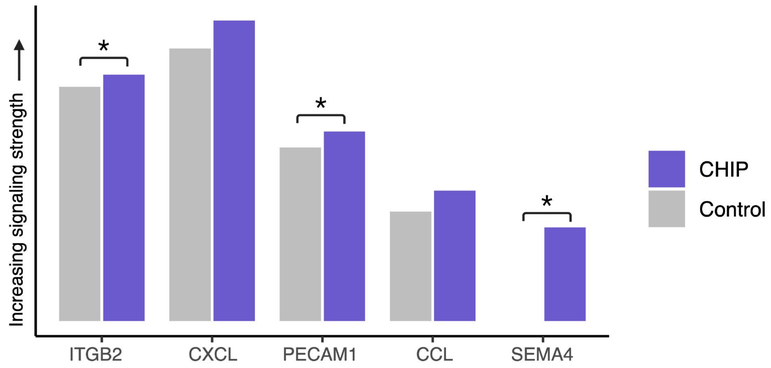


MONOCYTE-ENDOTHELIAL SIGNALING

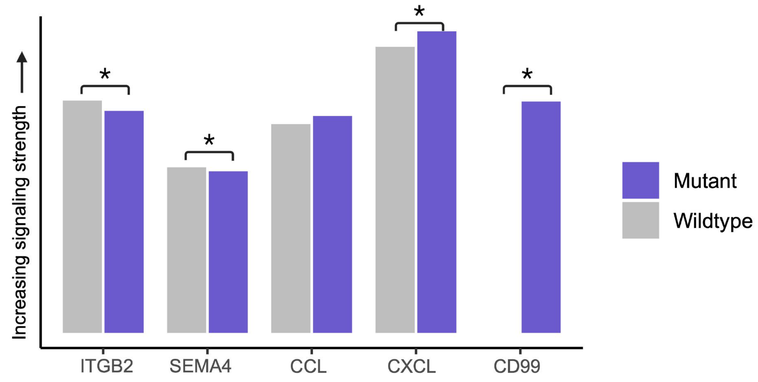
Monocyte-endothelial signaling patterns can be predicted using the R package CellChat[5].



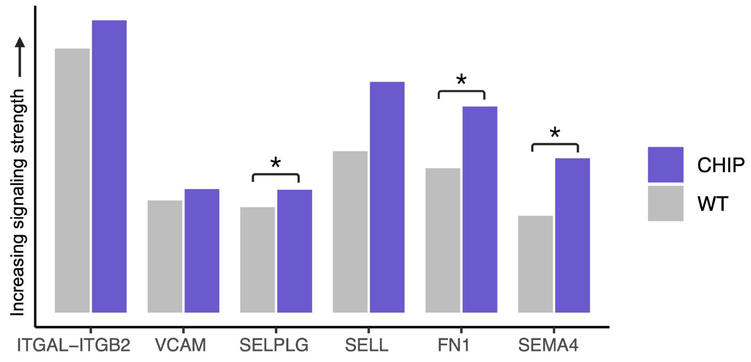
CHIP vs control monocyte to endothelial cell signaling
Cell adhesion signals are increased in CHIP.



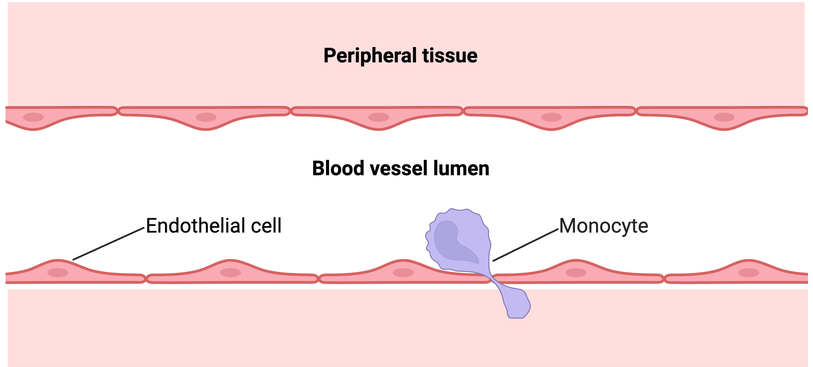
Mutant vs wildtype monocyte to endothelial cell signaling
Some cell adhesion signals are higher in mutant cells.



Mouse CHIP vs WT monocyte to endothelial cell signaling
Cell adhesion signals are increased with *TET2* knockout.



VASCULAR DISEASE PROGRESSION

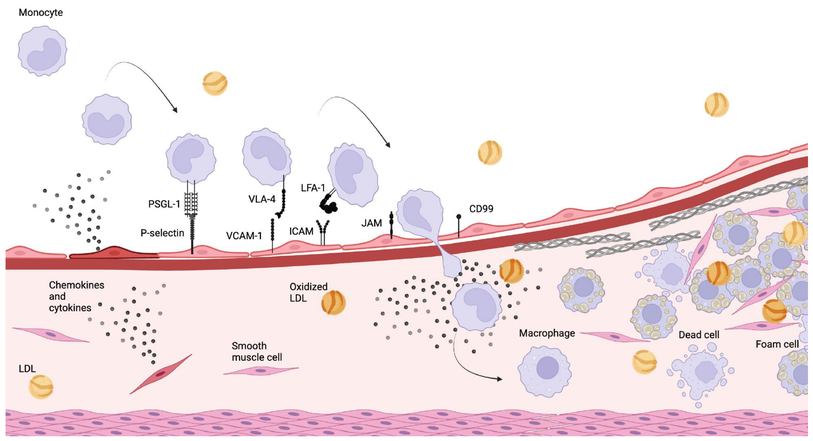


Dysregulation of transendothelial migration contributes to the pathogenesis of vascular diseases and increases risk for adverse cardiovascular outcomes[6].

Altered monocyte-endothelial interactions may explain vascular risks of CHIP.

Transendothelial migration is regulated by...

1. **Selectin**-dependent tethering and rolling
2. **Integrin**-dependent adhesion
3. **Adhesion molecule**-dependent migration[7]



CONCLUSIONS

CHIP mutant cells can be identified with mitochondrial markers.

Transendothelial migration signals are enhanced in CHIP.

Mutant and wildtype cells contribute to signaling increases.

FUNDING AND REFERENCES

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